

Benchmark of comparative single-cell analysis approaches

Group-Project SSBI 2026

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September 2026

Introduction

Analysing single-cell data brings some particular difficulties compared to bulk measurements: cells from the same donor are not independent replicates of that donor’s biology, and only a subset of a donor’s cells may show a phenotype of interest, so donor-level labels cannot simply be treated as cell-level labels. In this project we compare and try to improve CellCnn [1], an unsupervised representation-learning method developed to detect disease-associated cell subsets.

Our data come from Horowitz et al. [2] and cover 20 donors (9 CMV+, 11 CMV−), including 4 identifiable monozygotic twin pairs. Each donor’s history of cytomegalovirus (CMV) infection was recorded, since CMV infection is known to leave behind a population of long-lived, memory-like ("adaptive") natural killer (NK) cells. This CMV label was assigned donor-wise, even though only a small subpopulation of that donor’s NK cells actually carries the phenotypic signature linked to the infection. From each sample, 37 protein markers were measured by CyTOF, a technique that reads out metal-isotope-tagged antibodies via time-of-flight mass spectrometry. Before subsampling, NK cells made up between 2.4 % and 16.2 % of alive events per donor.

To avoid a bias arising from donors contributing very different numbers of cells, we randomly selected 2,000 cells from each donor to obtain a balanced subsample. To check representativeness, we compared the fraction of NKG2C⁺/CD57⁺ cells (a marker combination associated with the adaptive, CMV-associated NK phenotype) in the balanced subsample against the fraction in the complete cell population of each donor (Supp. Fig. 2): the subsample is representative for this specific two-marker phenotype, though we did not check other marker combinations or rarer populations.

Before dimensionality reduction, we applied an arcsinh transformation with a cofactor of 5, the standard default for CyTOF data [3, 4], which we validated visually via clean bimodal separation of known positive/negative markers (CD3, CD56, CD16, CD57) relative to smaller and larger candidate cofactors. We then z-scored each marker to give every marker equal weight in the distance-based methods below.

Dimensionality reduction

We started with principal component analysis (PCA). As shown in Fig. 1, cumulative explained variance grows roughly like a square-root function of the number of components, so the first two PCs together explain only 20.29 % of the total variance (Supp. Fig. 3). We use the first 29 PCs (91.5 % of the variance) as input to the non-linear methods below, both to reduce dimensionality before the more expensive computations and to average out marker-specific noise that concentrates in the low-variance trailing components.

Next, we swept the perplexity parameter of t-SNE [5] (via scikit-learn [6]) over the values 5, 15, 30, 40, 50, 60, 70 and 100, computing trustworthiness and kNN-preservation on a fixed random subsample of 5,000 of our 40,000 balanced cells. trustworthiness penalizes embeddings that place points close together that were not close in the high-dimensional input, weighted by how far out of the true neighbourhood a false neighbour originally ranked [7]; kNN-preservation is stricter, measuring the exact overlap of each point’s k nearest neighbours between the two representations, with no partial credit for a near-miss. trustworthiness scores in the range 50–100 were very

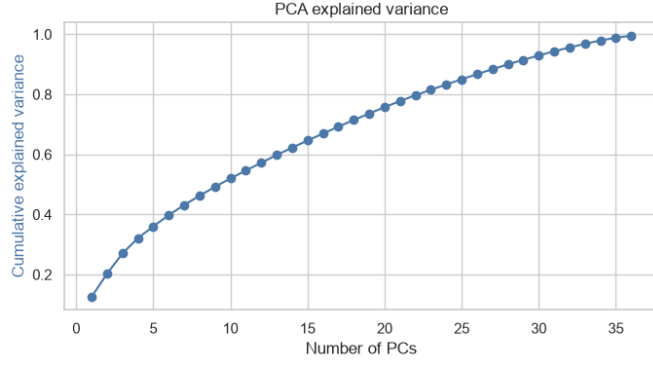


Figure 1: Cumulative explained variance (blue) across all 37 principal components.

similar (within about 0.0003), so we chose the marginal maximum, 60 (trustworthiness= 0.926). Visually, perplexity 5 gives a largely undifferentiated cloud, while higher perplexities show distinct clusters emerging (Supp. Fig. 4).

For UMAP [8] we grid-searched learning rate (0.1, 0.5, 1.0, 5.0), minimum distance (0.0, 0.1, 0.5, 1) and number of neighbours (5, 15, 50, 100). The highest trustworthiness was obtained at learning rate 0.1, 5 neighbours and minimum distance 0.0 (Table 3, full grid in Supp. Fig. 5). Visibly lower minimum distance packs points more tightly and the number of neighbours changes cluster shape but has a smaller effect than minimum distance.

We compare these three methods using trustworthiness, kNN-preservation and the Pearson correlation (r_d) of all pairwise distances. While the first two measures capture *local* structure (is each point’s immediate neighbourhood preserved), r_d captures *global* structure (is the overall arrangement of near and far points preserved, independent of exact neighbour identity). This distinction shows in the results (Table 1): trustworthiness and kNN-preservation agree on t-SNE as the best method, but r_d favours PCA instead, consistent with PCA being a linear projection explicitly built to preserve variance/distance.

Embedding	$T(15) \uparrow$	$R(15) \uparrow$	$r_d \uparrow$
PCA	0.7258	0.0072	0.6204
t-SNE	0.9612	0.2232	0.4868
UMAP	0.9009	0.1126	0.3954

Table 1: PCA, t-SNE and UMAP: local structure via trustworthiness (T) and kNN-preservation (R), global structure via Pearson correlation (r_d) of pairwise distances. T, R use $k = 15$. Green: best per column; red: worst.

Pair	$R(15) \uparrow$	$M^2 \downarrow$
PCA vs t-SNE	0.00585	0.45189
PCA vs UMAP	0.00410	0.73390
t-SNE vs UMAP	0.12532	0.72369

Table 2: Pairwise kNN-preservation (R , local) and Procrustes disparity (M^2 , aligned global shape).

Finally, we compare each pair of methods using kNN-preservation (R) and Procrustes disparity (M^2) (Table 2). Procrustes disparity aligns two maps by translation/rotation/reflection/scaling, then measures the remaining mismatch after this optimal alignment; a lower score means a closer overall shape once cosmetic differences in orientation and scale have been removed [9]. By this measure, PCA and t-SNE are the closest match, plausibly because our t-SNE runs were PCA-initialized, seeding its global layout before local optimization reshapes fine structure. By contrast, for local-neighbour agreement (R), t-SNE and UMAP are the more similar pair. The absolute R values look small, but chance level for $n = 40,000$, $k = 15$ is only $k/n \approx 0.0004$, so even the smallest value is well above chance.

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Supplementary Figures and Tables

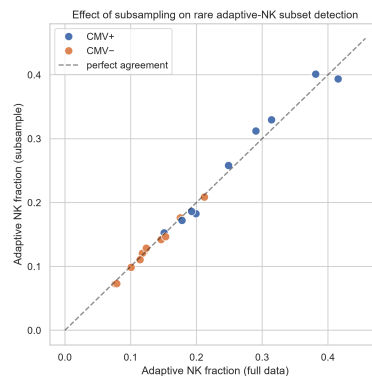


Figure 2: NKG2C⁺/CD57⁺ fraction, 2,000-cell subsample per donor vs. full population.

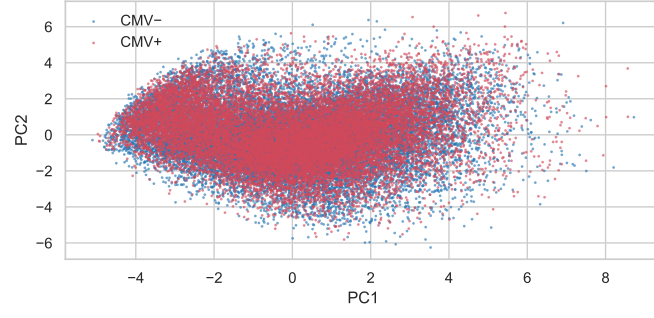


Figure 3: First two PCs, 40,000-cell sample, coloured by CMV status (CMV+: red, CMV-: blue).

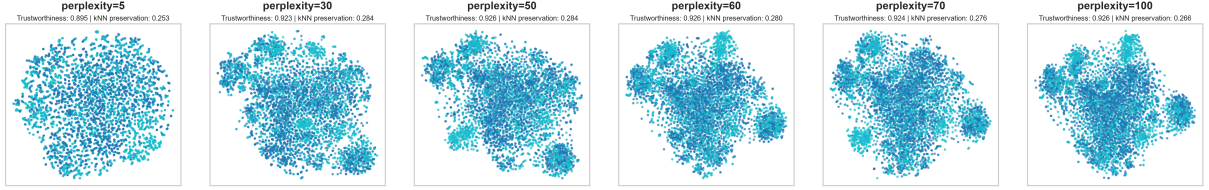


Figure 4: t-SNE embeddings, perplexities 5, 30, 50, 60, 70, 100; trustworthiness/kNN-preservation above each panel.



Figure 5: UMAP grid: effect of $n_{\text{neighbors}}$ and min_dist at learning rate 0.1.

learning_rate	$n_{\text{neighbors}}$	min_dist	$T(15) \uparrow$	$R(15) \uparrow$
0.1	5	0.0	0.895908	0.230373
0.5	5	0.0	0.893360	0.226933
1.0	5	0.0	0.890139	0.224227
5.0	5	0.0	0.878033	0.203747
0.1	50	0.1	0.864503	0.149040
0.5	50	0.1	0.862615	0.143093
1.0	50	0.1	0.856671	0.135920
5.0	50	0.1	0.840111	0.111320

Table 3: Effect of UMAP learning rate on trustworthiness/kNN-preservation for two fixed ($n_{\text{neighbors}}$, min_dist) settings.